1.55 Equations of Lines in 3D - Cartesian Equations

Worked Solutions

1 a)

$$\frac{x-4}{2} = \frac{y-1}{3} = \frac{z+2}{8}.$$

1 b)

$$\frac{x-2}{9} = \frac{y+8}{-1} = \frac{z+5}{-2}.$$

1 c)

$$\frac{x+3}{2} = \frac{y+5}{-1} = \frac{z-6}{-5} \quad \text{or} \quad \frac{x+3}{2} = \frac{y+5}{-1} = \frac{6-z}{5}.$$

1 d)

$$\frac{x-1}{-1} = \frac{y+1}{3} = \frac{z-0}{10} \quad \text{or} \quad 1-x = \frac{y+1}{3} = \frac{z}{10}.$$

1 e)

$$\frac{x-0}{-1} = \frac{y-8}{-5} = \frac{z+3}{9} \quad \text{or} \quad -x = \frac{8-y}{5} = \frac{z+3}{9}.$$

1 f)

$$\frac{x-\frac{1}{2}}{5} = \frac{y+3}{-2} = \frac{z+\frac{1}{3}}{-10} \quad \text{or} \quad \frac{2x-1}{10} = \frac{y+3}{-2} = \frac{3z+1}{-30}.$$

1 g)

$$\frac{x+\frac{1}{4}}{10} = \frac{y-\frac{1}{2}}{-4} = \frac{z-\frac{2}{3}}{-3} \quad \text{or} \quad \frac{4x+1}{40} = \frac{1-2y}{8} = \frac{2-3z}{9}.$$

1 h)

$$\frac{x-2}{3} = \frac{y+5}{4}, \quad z = -7.$$

1 i)

$$\frac{x+7}{8} = \frac{z-12}{-3}, \quad y = 3 \quad \text{or} \quad \frac{x+7}{8} = \frac{12-z}{3}, \quad y = 3.$$

1 j)

$$x = -4, \quad \frac{y-3}{-5} = \frac{z-3}{6} \quad \text{or} \quad x = -4, \quad \frac{3-y}{5} = \frac{z-3}{6}.$$

1 k)

$y = 3, \quad z = -7.$

1 l)

$$\frac{x-\frac{3}{4}}{-16} = \frac{z-\frac{9}{4}}{\frac{3}{4}}, \quad y = -\frac{5}{3} \quad \text{or} \quad \frac{3-4x}{64} = \frac{4z-9}{12}, \quad y = -\frac{5}{3}.$$

2)

When $z = 6$:

$$\frac{x-1}{3} = \frac{y+2}{4} = \frac{6-5}{2} = \frac{1}{2}.$$
$$\frac{x-1}{3} = \frac{1}{2} \Rightarrow x = \frac{5}{2}.$$
$$\frac{y+2}{4} = \frac{1}{2} \Rightarrow y = 0.$$

So the point has coordinates $\left(\frac{5}{2}, 0, 6\right)$.

3)

When $y = 8$:

$$\frac{x+5}{6} = \frac{z+8}{-3} = \frac{2-8}{10} = -\frac{3}{5}.$$
$$\frac{x+5}{6} = -\frac{3}{5} \Rightarrow x = -\frac{43}{5}.$$
$$\frac{z+8}{-3} = -\frac{3}{5} \Rightarrow z = -\frac{31}{5}.$$

So the point has coordinates $\left(-\frac{43}{5}, 8, -\frac{31}{5}\right)$.

4)

When $x = -5$:

$$\frac{-5-3}{-2} = 4.$$

When $y = 20$:

$$\frac{20-4}{4} = 4.$$

When $z = 6$:

$$\frac{6+6}{3} = 4.$$

The values are consistent, so $(-5, 20, 6)$ lies on the line.

5)

When $x = 2$:

$$\frac{2-9}{3} = -\frac{7}{3}.$$

When $y = -8$:

$$\frac{-8-13}{9} = -\frac{21}{9} = -\frac{7}{3}.$$

The values are consistent, so $(2, -8, 3)$ lies on the line.

6)

When $x = 15$:

$$\frac{15+2}{8} = \frac{17}{8}.$$

When $y = 11$:

$$\frac{11+57}{32} = \frac{68}{32} = \frac{17}{8}.$$

When $z = -10$:

$$\frac{-10+18}{4} = \frac{8}{4} = 2.$$

The values are not consistent, so $(15, 11, -10)$ does not lie on the line.

7 a)

$$\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}.$$

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{6}.$$

7 b)

$$\begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}.$$

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{-3}.$$

7 c)

$$\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -8 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 8 \end{pmatrix}.$$

$$\frac{x-4}{1} = \frac{y}{1} = \frac{z+8}{8} \quad \text{or} \quad x - 4 = y = \frac{z+8}{8}.$$

7 d)

$$\begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -8 \\ 3 \end{pmatrix}.$$

$$x = 2, \quad \frac{y-5}{-8} = \frac{z+1}{3}.$$

7 e)

$$\begin{pmatrix} 9 \\ 8 \\ -6 \end{pmatrix} - \begin{pmatrix} 3 \\ 8 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ -7 \end{pmatrix}.$$

$$\frac{x-3}{6} = \frac{z-1}{-7}, \quad y = 8.$$

7 f)

$$\begin{pmatrix} 6 \\ -7 \\ -2 \end{pmatrix} - \begin{pmatrix} 5 \\ -3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix}.$$

$$\frac{x-5}{1} = \frac{y+3}{-4}, \quad z = -2 \quad \text{or} \quad x - 5 = \frac{y+3}{-4}, \quad z = -2.$$

8)

Rewrite as:

$$l_1 : \frac{x-\frac{3}{2}}{3} = \frac{y+\frac{11}{5}}{-\frac{4}{5}} = \frac{z-\frac{9}{8}}{\frac{3}{2}}.$$

$$l_2 : \frac{3x-1}{-9} = \frac{10y-13}{8} = \frac{2z-5}{-3} \Rightarrow \frac{x-\frac{1}{3}}{-3} = \frac{y-\frac{13}{10}}{\frac{4}{5}} = \frac{z-\frac{5}{2}}{-\frac{3}{2}}.$$

The lines are parallel if there exists a k such that

$$\begin{pmatrix} 3 \\ -\frac{4}{5} \\ \frac{3}{2} \end{pmatrix} = k \begin{pmatrix} -3 \\ \frac{4}{5} \\ -\frac{3}{2} \end{pmatrix}.$$

$$3 = -3k \Rightarrow k = -1.$$

$$-\frac{4}{5} = \frac{4}{5}k \Rightarrow k = -1.$$

$$\frac{3}{2} = -\frac{3}{2}k \Rightarrow k = -1.$$

As the values of k are consistent, the lines are parallel.

9)

l_1 :

$$\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 74 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 10 \end{pmatrix}.$$

l_2 :

$$\mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 6 \\ 8 \end{pmatrix}.$$

$$3 + 2\lambda = -1 + 3\mu \Rightarrow 2\lambda - 3\mu = -4 \quad (1).$$

$$-2 + 3\lambda = 4 + 6\mu \Rightarrow 3\lambda - 6\mu = 6 \quad (2).$$

$$74 + 10\lambda = -2 + 8\mu \Rightarrow 10\lambda - 8\mu = -76 \quad (3).$$

Solve (1) and (2): $\lambda = -14$ and $\mu = -8$. Check in (3):

$$10(-14) - 8(-8) = -76 \quad \text{as required.}$$

So the lines intersect. Substitute $\lambda = -14$ into l_1: The lines intersect at $(-25, -44, -66)$.

10)

Rewrite l_1 as

$$\frac{x+5}{-\frac{8}{5}} = \frac{y-5}{\frac{2}{5}} = \frac{z+\frac{3}{5}}{-\frac{4}{5}}.$$

Rewrite l_2 as

$$\frac{x+4}{-\frac{3}{5}} = \frac{y+5}{-\frac{3}{5}} = \frac{z-9}{\frac{2}{5}}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -5 \\ 5 \\ -\frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} -\frac{8}{5} \\ \frac{2}{5} \\ -\frac{4}{5} \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} -5 \\ 5 \\ -\frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 1 \\ -2 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ -5 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} -\frac{3}{5} \\ -\frac{3}{5} \\ \frac{2}{5} \end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ -5 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -3 \\ 2 \end{pmatrix}.$$

$$-5 - 4\lambda = -4 - 3\mu \Rightarrow -4\lambda + 3\mu = 1 \quad (1).$$

$$5 + \lambda = -5 - 3\mu \Rightarrow \lambda + 3\mu = -10 \quad (2).$$

$$-\frac{3}{5} - 2\lambda = 9 + 2\mu \Rightarrow -2\lambda - 2\mu = \frac{48}{5} \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{11}{5}$ and $\mu = -\frac{13}{5}$. Check in (3):

$$-2\left(-\frac{11}{5}\right) - 2\left(-\frac{13}{5}\right) = \frac{48}{5} \quad \text{as required.}$$

So the lines intersect. Substitute $\lambda = -\frac{11}{5}$ into l_1: The lines intersect at $\left(\frac{19}{5}, \frac{14}{5}, \frac{19}{5}\right)$.

11 a)

Rewrite as l_1:

$$\frac{x+1}{4} = \frac{y-3}{-5} = \frac{z-7}{-2}.$$

Rewrite as l_2:

$$\frac{x-9}{-4} = \frac{y+2}{5} = \frac{z+1}{2}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ \frac{7}{2} \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -5 \\ -2 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 9 \\ -2 \\ -\frac{1}{3} \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 5 \\ 2 \end{pmatrix}.$$

$$4 = k(-4) \Rightarrow k = -1.$$

$$-5 = k(5) \Rightarrow k = -1.$$

$$-2 = k(2) \Rightarrow k = -1.$$

Consistent values of k, so the lines are parallel (or coincident). Check if $\left(9, -2, -\frac{1}{3}\right)$ lies on l_1:

$$9 = -1 + 4\lambda \Rightarrow \lambda = \frac{5}{2}.$$

$$-2 = 3 - 5\lambda \Rightarrow \lambda = 1.$$

$$-\frac{1}{3} = \frac{7}{2} - 2\lambda \Rightarrow \lambda = \frac{23}{12}.$$

Inconsistent values of λ, so the lines are not coincident. So the lines are parallel.

11 b)

Rewrite as l_1:

$$\frac{x+2}{3} = \frac{y-1}{\frac{3}{4}} = \frac{z-2}{-1}.$$

Rewrite as l_2:

$$\frac{x+3}{3} = \frac{y+3}{\frac{8}{3}} = \frac{z+3}{1}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ \frac{1}{4} \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ \frac{3}{4} \\ -1 \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ \frac{1}{4} \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 12 \\ 3 \\ -4 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -\frac{3}{8} \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ \frac{3}{4} \\ 1 \end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -\frac{3}{8} \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 12 \\ 3 \\ 4 \end{pmatrix}.$$

$$12 = k(12) \Rightarrow k = 1.$$

$$3 = k(3) \Rightarrow k = 1.$$

$$-4 = k(4) \Rightarrow k = -1.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$-2 + 12\lambda = -3 + 12\mu \Rightarrow 12\lambda - 12\mu = -1 \quad (1).$$

$$\frac{1}{4} + 3\lambda = -\frac{3}{8} + 3\mu \Rightarrow 3\lambda - 3\mu = -\frac{5}{8} \quad (2).$$

$$2 - 4\lambda = -3 + 4\mu \Rightarrow -4\lambda - 4\mu = -5 \quad (3).$$

Solve (2) and (3): $\lambda = \frac{25}{48}$ and $\mu = \frac{35}{48}$. Check in (1):

$$12\left(\frac{25}{48}\right) - 12\left(\frac{35}{48}\right) = -\frac{5}{2} \neq -1.$$

So the lines do not intersect. l_1 and l_2 are skew lines.

11 c)

Rewrite as l_1:

$$\frac{x-1}{\frac{1}{2}} = \frac{y+7}{1} = \frac{z-3}{-8}.$$

Rewrite as l_2:

$$\frac{x-7}{\frac{1}{2}} = \frac{y+1}{2} = \frac{z-3}{1}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} \frac{1}{2} \\ 1 \\ -8 \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -16 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} \frac{7}{6} \\ -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} + \mu \begin{pmatrix} \frac{1}{2} \\ 2 \\ 1 \end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r} = \begin{pmatrix} \frac{7}{6} \\ -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}.$$

$$1 = k(1) \Rightarrow k = 1.$$

$$2 = k(4) \Rightarrow k = \frac{1}{2}.$$

$$-16 = k(2) \Rightarrow k = -8.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$1 + \lambda = \frac{7}{6} + \mu \Rightarrow \lambda - \mu = \frac{11}{12} \quad (1).$$

$$-7 + 2\lambda = -\frac{1}{2} + 4\mu \Rightarrow 2\lambda - 4\mu = \frac{13}{2} \quad (2).$$

$$3 - 16\lambda = \frac{3}{2} + 2\mu \Rightarrow -16\lambda - 2\mu = -\frac{3}{2} \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{17}{12}$ and $\mu = -\frac{7}{3}$. Check in (3):

$$-16\left(-\frac{17}{12}\right) - 2\left(-\frac{7}{3}\right) = \frac{82}{3} \neq -\frac{3}{2}.$$

So the lines do not intersect. l_1 and l_2 are skew lines.

11 d)

Rewrite as l_1:

$$\frac{4x-1}{-8} = \frac{y+5}{3} = \frac{5z-65}{4} \quad \text{and then as} \quad l_1 : \frac{x-\frac{1}{4}}{-2} = \frac{y+5}{3} = \frac{z-13}{\frac{4}{5}}.$$

Rewrite as l_2:

$$\frac{3x-2}{5} = \frac{4y-7}{-9} = \frac{z-3}{-1} \quad \text{and then as} \quad l_2 : \frac{x-\frac{2}{3}}{\frac{5}{3}} = \frac{y-\frac{7}{4}}{-\frac{9}{4}} = \frac{z-3}{-1}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -5 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ \frac{4}{5} \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ -5 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -10 \\ 15 \\ 4 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} \frac{2}{3} \\ \frac{7}{4} \\ 3 \end{pmatrix} + \mu \begin{pmatrix} \frac{5}{3} \\ -\frac{9}{4} \\ -1 \end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r} = \begin{pmatrix} \frac{2}{3} \\ \frac{7}{4} \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 20 \\ -27 \\ -12 \end{pmatrix}.$$

$$-10 = k(20) \Rightarrow k = -\frac{1}{2}.$$

$$15 = k(-27) \Rightarrow k = -\frac{5}{9}.$$

$$4 = k(-12) \Rightarrow k = -\frac{1}{3}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$\frac{1}{4} - 10\lambda = \frac{2}{3} + 20\mu \Rightarrow -10\lambda - 20\mu = \frac{5}{12} \quad (1).$$

$$-5 + 15\lambda = \frac{7}{4} - 27\mu \Rightarrow 15\lambda + 27\mu = \frac{27}{4} \quad (2).$$

$$13 + 4\lambda = 3 - 12\mu \Rightarrow 4\lambda + 12\mu = -10 \quad (3).$$

Solve (1) and (2): $\lambda = \frac{39}{8}$ and $\mu = -\frac{59}{24}$. Check in (3):

$$4\left(\frac{39}{8}\right) + 12\left(-\frac{59}{24}\right) = -10 \quad \text{as required.}$$

So the lines do intersect. Substitute $\lambda = \frac{39}{8}$ into l_1 to find where the lines intersect:

$$\left(\frac{1}{4} + \left(\frac{39}{8}\right) \times (-10), -5 + \left(\frac{39}{8}\right) \times 15, 13 + \left(\frac{39}{8}\right) \times 4\right) = \left(-\frac{97}{2}, \frac{545}{8}, \frac{65}{2}\right).$$

12 a)

$$l_1 : \mathbf{r} = \begin{pmatrix} 6 \\ -1 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 15 \\ -3 \\ -12 \end{pmatrix} + \mu \begin{pmatrix} k \\ 9 \\ 6 \end{pmatrix}.$$

$$6 - 3\lambda = 15 + k\mu \Rightarrow -3\lambda - k\mu = 9 \quad (1).$$

$$-1 + 2\lambda = -3 + 9\mu \Rightarrow 2\lambda - 9\mu = -2 \quad (2).$$

$$7 + 4\lambda = -12 + 6\mu \Rightarrow 4\lambda - 6\mu = -19 \quad (3).$$

Solve (2) and (3): $\lambda = -\frac{53}{8}$ and $\mu = -\frac{5}{4}$. Substitute into (1):

$$-3\left(-\frac{53}{8}\right) - k\left(-\frac{5}{4}\right) = 9.$$

$$\Rightarrow \frac{159}{8} + \frac{5}{4}k = 9.$$

$$\Rightarrow k = -\frac{87}{10}.$$

12 b)

Substitute $\lambda = -\frac{53}{8}$ into l_1 to find where the lines intersect:

$$\left(6+\left(-\frac{53}{8}\right)\times(-3), -1+\left(-\frac{53}{8}\right)\times 2, 7+\left(-\frac{53}{8}\right)\times 4\right) = \left(\frac{207}{8}, -\frac{57}{4}, -\frac{39}{2}\right).$$

The position vector of the point of intersection is

$$\begin{pmatrix} \frac{207}{8} \\ -\frac{57}{4} \\ -\frac{39}{2} \end{pmatrix}.$$

13 a)

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -7 \\ 1 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}.$$

$$\cos\theta = \frac{\begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}}{\left|\begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}\right| \left|\begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}\right|} = \frac{2\times 4 + 9\times 2 + 4\times 3}{\sqrt{2^2+9^2+4^2}\sqrt{4^2+2^2+3^2}} = \frac{38}{\sqrt{101}\sqrt{29}} = \frac{38}{\sqrt{2929}}.$$

$$\theta = \arccos\left(\frac{38}{\sqrt{2929}}\right) = 45.4^\circ \text{ (1 DP)}.$$

13 b)

Rewrite as l_1:

$$\frac{x+2}{-8} = \frac{y-1}{2} = z+2 \quad \text{and then as} \quad \frac{x+2}{-8} = \frac{y-1}{2} = \frac{z+2}{1}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}.$$

$$\cos\theta = \frac{\begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}}{\left|\begin{pmatrix} -8 \\ 2 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}\right|} = \frac{(-8)\times 4 + 2\times 3 + 1\times 2}{\sqrt{(-8)^2+2^2+1^2}\sqrt{4^2+3^2+2^2}} = \frac{-24}{\sqrt{69}\sqrt{29}} = -\frac{8\sqrt{2001}}{667}.$$

$$\theta = \arccos\left(-\frac{8\sqrt{2001}}{667}\right) = 122.447\ldots^\circ.$$

Acute Angle $= 180^\circ - 122.447\ldots^\circ = 57.6^\circ$ (1 DP).

13 c)

Rewrite as l_1:

$$\frac{3-x}{2} = \frac{y+1}{-3} = 3-z \quad \text{and then as} \quad \frac{x-3}{-2} = \frac{y+1}{-3} = \frac{z-3}{-1}.$$

Rewrite as l_2: $x+2 = 3-y = 4-z$ and then as

$$\frac{x+2}{1} = \frac{y-3}{-1} = \frac{z-4}{-1}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}.$$

$$\cos\theta = \frac{\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}}{\left|\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}\right|} = \frac{2\times 1 + 3\times(-1) + 1\times(-1)}{\sqrt{2^2+3^2+1^2}\sqrt{1^2+(-1)^2+(-1)^2}} = -\frac{2}{\sqrt{14}\sqrt{3}} = -\frac{\sqrt{42}}{21}.$$

$$\theta = \arccos\left(-\frac{\sqrt{42}}{21}\right) = 107.975\ldots^\circ.$$

Acute Angle $= 180^\circ - 107.975\ldots^\circ = 72.0^\circ$ (1 DP).

13 d)

Rewrite as l_1: $x = 3, \quad \frac{2-y}{2} = z+8$ and then as

$$l_1 : x = 3, \quad \frac{y-2}{-2} = \frac{z+8}{1}.$$

Rewrite as l_2: $x = 8, \quad y-9 = \frac{z-4}{5}$ and then as

$$l_2 : x = 8, \quad \frac{y-9}{1} = \frac{z-4}{5}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ 9 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}.$$

$$\cos\theta = \frac{\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}}{\left|\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}\right|} = \frac{0\times 0 + (-2)\times 1 + 1\times 5}{\sqrt{0^2+(-2)^2+1^2}\sqrt{0^2+1^2+5^2}} = \frac{3}{\sqrt{5}\sqrt{26}} = \frac{3\sqrt{130}}{130}.$$

$$\theta = \arccos\left(\frac{3\sqrt{130}}{130}\right) = 74.7^\circ \text{ (1 DP)}.$$

13 e)

Rewrite as l_1:

$$\frac{2x-3}{4}=\frac{3y-7}{-6}=\frac{2z+1}{4} \quad \text{and then as} \quad \frac{x-\frac{3}{2}}{2}=\frac{y-\frac{7}{3}}{-2}=\frac{z+\frac{1}{2}}{2}.$$

Rewrite as l_2:

$$\frac{2x+1}{3}=\frac{4y-6}{-5}=\frac{z-10}{-1} \quad \text{and then as} \quad \frac{x+\frac{1}{2}}{\frac{3}{2}}=\frac{y-\frac{3}{2}}{-\frac{5}{4}}=\frac{z-10}{-1}.$$

$$l_1 : \mathbf{r}=\begin{pmatrix}\frac{3}{2}\\ \frac{7}{3}\\ -\frac{1}{2}\end{pmatrix}+\lambda\begin{pmatrix}2\\-2\\2\end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r}=\begin{pmatrix}\frac{3}{2}\\ \frac{7}{3}\\ -\frac{1}{2}\end{pmatrix}+\lambda\begin{pmatrix}1\\-1\\1\end{pmatrix}.$$

$$l_2 : \mathbf{r}=\begin{pmatrix}-\frac{1}{2}\\ \frac{3}{2}\\ 10\end{pmatrix}+\mu\begin{pmatrix}\frac{3}{2}\\ -\frac{5}{4}\\ -1\end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r}=\begin{pmatrix}-\frac{1}{2}\\ \frac{3}{2}\\ 10\end{pmatrix}+\mu\begin{pmatrix}6\\-5\\-4\end{pmatrix}.$$

$$\cos\theta=\frac{\begin{pmatrix}1\\-1\\1\end{pmatrix}\cdot\begin{pmatrix}6\\-5\\-4\end{pmatrix}}{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\right|\left|\begin{pmatrix}6\\-5\\-4\end{pmatrix}\right|}=\frac{1\times 6+(-1)\times(-5)+1\times(-4)}{\sqrt{1^2+(-1)^2+1^2}\sqrt{6^2+(-5)^2+(-4)^2}}=\frac{7}{\sqrt{3}\sqrt{77}}=\frac{\sqrt{231}}{33}.$$

$$\theta=\arccos\left(\frac{\sqrt{231}}{33}\right)=62.6^\circ \text{ (1 DP)}.$$

13 f)

Rewrite as l_1:

$$\frac{x-5}{-1}=\frac{y-9}{4}, \quad z=5.$$

Rewrite as l_2:

$$\frac{8x+3}{3}=\frac{z-4}{5}, \quad y=-2 \quad \text{and then as} \quad \frac{x+\frac{3}{8}}{\frac{3}{8}}=\frac{z-4}{5}, \quad y=-2.$$

$$l_1 : \mathbf{r}=\begin{pmatrix}5\\9\\5\end{pmatrix}+\lambda\begin{pmatrix}-1\\4\\0\end{pmatrix}.$$

$$l_2 : \mathbf{r}=\begin{pmatrix}-\frac{3}{8}\\-2\\4\end{pmatrix}+\mu\begin{pmatrix}\frac{3}{8}\\0\\5\end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r}=\begin{pmatrix}-\frac{3}{8}\\-2\\4\end{pmatrix}+\mu\begin{pmatrix}3\\0\\40\end{pmatrix}.$$

$$\cos\theta=\frac{\begin{pmatrix}-1\\4\\0\end{pmatrix}\cdot\begin{pmatrix}3\\0\\40\end{pmatrix}}{\left|\begin{pmatrix}-1\\4\\0\end{pmatrix}\right|\left|\begin{pmatrix}3\\0\\40\end{pmatrix}\right|}=\frac{(-1)\times 3+4\times 0+0\times 40}{\sqrt{(-1)^2+4^2+0^2}\sqrt{3^2+0^2+40^2}}=-\frac{3}{\sqrt{17}\sqrt{1609}}=-\frac{3}{\sqrt{27353}}.$$

$$\theta=\arccos\left(-\frac{3}{\sqrt{27353}}\right)=91.039\ldots^\circ.$$

Acute angle is $180^\circ-91.039\ldots^\circ=89.0^\circ$ (1 DP).

14 a)

Rewrite as l_2:

$$\frac{x+200}{15k} = \frac{y-380}{18} = \frac{z-475}{-5}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 2140 \\ 425 \\ 670 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 9 \\ 0 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -200 \\ 380 \\ 475 \end{pmatrix} + \mu \begin{pmatrix} 15k \\ 18 \\ -5 \end{pmatrix}.$$

$$2140 + 10\lambda = -200 + 15k\mu \Rightarrow 10\lambda - 15k\mu = -2340 \quad (1).$$

$$425 + 9\lambda = 380 + 18\mu \Rightarrow 9\lambda - 18\mu = -45 \quad (2).$$

$$670 = 475 - 5\mu \Rightarrow \mu = -39 \quad (3).$$

Solve (2) and (3): $\lambda = -83$ and $\mu = -39$. Substitute into (1):

$$10(-83) - 15k(-39) = -2340.$$

$$\Rightarrow -830 + 585k = -2340.$$

$$\Rightarrow k = -\frac{302}{117}.$$

14 b)

Substitute $\lambda = -83$ into l_1 to find where the lines intersect:

$$(2140 + (-83) \times 10, 425 + (-83) \times 9, 670) = (1310, -322, 670).$$

14 c)

The asteroids may be travelling at different speeds.

15 a)

Rewrite as l_1:

$$\frac{x+1}{4} = \frac{y-3}{-1} = \frac{3z+4}{6} \quad \text{and then as} \quad \frac{x+1}{4} = \frac{y-3}{-1} = \frac{z+\frac{4}{3}}{2}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -\frac{4}{3} \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}.$$

$$\begin{pmatrix} 6 \\ k \\ k+1 \end{pmatrix} - \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix}.$$

l_1 and l_2 are perpendicular so:

$$\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ k \\ k-3 \end{pmatrix} = 0.$$

$$\Rightarrow 4 \times 1 + (-1) \times k + 2 \times (k-3) = 0.$$
$$\Rightarrow 4 - k + 2k - 6 = 0.$$
$$\Rightarrow k - 2 = 0 \Rightarrow k = 2.$$

15 b)

$$\begin{pmatrix} 6 \\ k \\ k-1 \end{pmatrix} - \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ k \\ k-5 \end{pmatrix}.$$

As $k = 2$, the direction vector of l_3 is $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$.

$$l_3 : \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}.$$

$$\cos\theta = \frac{\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}}{\left|\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}\right| \left|\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}\right|} = \frac{4 \times 1 + (-1) \times 2 + 2 \times (-3)}{\sqrt{4^2 + (-1)^2 + 2^2}\sqrt{1^2 + 2^2 + (-3)^2}}.$$

$$= -\frac{4}{\sqrt{21}\sqrt{14}} = -\frac{4}{\sqrt{294}} = -\frac{4}{7\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = -\frac{4\sqrt{6}}{42} = -\frac{2\sqrt{6}}{21}.$$

16 a)

Rewrite as l_1:

$$\frac{x-5}{-2} = \frac{y-3}{-1} = \frac{2z}{1} \quad \text{and then as} \quad \frac{x-5}{-2} = \frac{y-3}{-1} = \frac{z}{\frac{1}{2}}.$$

Rewrite as l_2:

$$\frac{x-2}{8-k} = \frac{y+5}{2p+5} = \frac{z-6}{-4}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ \frac{1}{2} \end{pmatrix} \quad \text{or} \quad l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}.$$

l_1 and l_2 are parallel:

$$\begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix} = q \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}.$$

$$-4 = q(8-k) \quad (1).$$

$$-2 = q(2p+5) \quad (2).$$

$$1 = q(-4) \quad (3).$$

Solve (3) to get $q = -\frac{1}{4}$. Substitute into (1):

$$-4 = -\frac{1}{4}(8-k) \Rightarrow k = -8.$$

Substitute into (2):

$$-2 = -\frac{1}{4}(2p+5) \Rightarrow p = \frac{3}{2}.$$

So $k = -8$ and $p = \frac{3}{2}$.

16 b)

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix}.$$

l_1 and l_2 intersect at right angles:

$$\begin{pmatrix} -4 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 8-k \\ 2p+5 \\ -4 \end{pmatrix} = 0.$$

$$\Rightarrow (-4) \times (8-k) + (-2) \times (2p+5) + 1 \times (-4) = 0.$$

$$\Rightarrow -32 + 4k - 4p - 10 - 4 = 0.$$

$$\Rightarrow 4k - 4p = 46.$$

$$\Rightarrow k = p + \frac{23}{2} \quad (1).$$

Now find where they intersect:

$$5 - 4\lambda = 2 + (8-k)\mu \quad (2).$$

$$3 - 2\lambda = -5 + (2p+5)\mu \quad (3).$$

$$\lambda = 6 - 4\mu \quad (4).$$

Substitute (1) and (4) into (2):

$$5 - 4(6 - 4\mu) = 2 + \left(8 - \left(p + \frac{23}{2}\right)\right)\mu.$$

$$\Rightarrow -19 + 16\mu = 2 - p\mu - \frac{7}{2}\mu.$$

$$\Rightarrow p\mu = -\frac{39}{2}\mu + 21.$$

$$\Rightarrow p = -\frac{39}{2} + \frac{21}{\mu} \quad (5).$$

Substitute (4) into (3):

$$3 - 2(6 - 4\mu) = -5 + (2p+5)\mu.$$

$$\Rightarrow -9 + 8\mu = -5 + 2p\mu + 5\mu.$$

$$\Rightarrow 2p\mu = -4 + 3\mu.$$

$$\Rightarrow p = -\frac{2}{\mu} + \frac{3}{2} \quad (6).$$

Equating (5) and (6):

$$-\frac{39}{2} + \frac{21}{\mu} = -\frac{2}{\mu} + \frac{3}{2}.$$
$$\Rightarrow -39\mu + 42 = -4 + 3\mu.$$
$$\Rightarrow 42\mu - 46 = 0.$$
$$\Rightarrow \mu = \frac{23}{21}.$$

Then $p = -\frac{2}{(23/21)} + \frac{3}{2} = -\frac{15}{46}$. And $k = (-\frac{15}{46}) + \frac{23}{2} = \frac{257}{23}$.

17)

Rewrite as l_1:

$$\frac{x-3}{7} = \frac{2y+1}{2} = \frac{z-1}{-1} \quad \text{and then as} \quad \frac{x-3}{7} = \frac{y+\frac{1}{2}}{1} = \frac{z-1}{-1}.$$

Rewrite as l_2:

$$\frac{x-8}{3} = \frac{y-\frac{21}{2}}{\frac{3}{2}} = \frac{z+4}{-\frac{3}{2}}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ \frac{21}{2} \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ \frac{3}{2} \\ -\frac{3}{2} \end{pmatrix} \quad \text{or} \quad l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ \frac{21}{2} \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

$$\Pi : \mathbf{r} = \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

18)

Rewrite as l_1:

$$\frac{x-3}{-1} = \frac{y-4}{5} = \frac{z+1}{1}.$$

Rewrite as l_2:

$$\frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+20}{6}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 2 \\ -20 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix}.$$

$$\begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 5 & 1 \\ 2 & -1 & 6 \end{vmatrix}.$$

$$= (5 \times 6 - 1 \times (-1))\mathbf{i} - ((-1) \times 6 - 1 \times 2)\mathbf{j} + ((-1) \times (-1) - 5 \times 2)\mathbf{k}.$$

$$= 31\mathbf{i} + 8\mathbf{j} - 9\mathbf{k}.$$

So $\Pi : 31x + 8y - 9z = d$. Substitute $(3, 4, -1)$ into the equation to find d:

$$31(3) + 8(4) - 9(-1) = d \Rightarrow d = 134.$$

$$\therefore \Pi : 31x + 8y - 9z = 134.$$

19 a)

$$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}.$$

19 b)

$$\mathbf{r} = \begin{pmatrix} 0 \\ -9 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 4 \\ -5 \end{pmatrix}.$$

19 c)

Rewrite as

$$\frac{x+8}{3} = \frac{y-2}{-3} = \frac{z}{10}.$$

$$\mathbf{r} = \begin{pmatrix} -8 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -3 \\ 10 \end{pmatrix}.$$

19 d)

Rewrite as

$$\frac{x-7}{-5} = \frac{y-8}{12} = \frac{z+3}{1}.$$

$$\mathbf{r} = \begin{pmatrix} 7 \\ 8 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 12 \\ 1 \end{pmatrix}.$$

19 e)

Rewrite as

$$\frac{x-\frac{3}{2}}{\frac{5}{2}} = \frac{y+\frac{1}{3}}{\frac{2}{3}} = \frac{z-\frac{3}{5}}{\frac{2}{5}}.$$

$$\mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{3} \\ \frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} \frac{5}{2} \\ \frac{2}{3} \\ \frac{2}{5} \end{pmatrix} \quad \text{or} \quad \mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{3} \\ \frac{3}{5} \end{pmatrix} + \lambda \begin{pmatrix} 75 \\ 20 \\ 12 \end{pmatrix}.$$

19 f)

Rewrite as

$$\frac{4x-3}{-6} = \frac{2y-5}{-2} = \frac{8z-6}{5}.$$

And then as

$$\frac{x-\frac{3}{4}}{-\frac{3}{2}} = \frac{y-\frac{5}{2}}{-1} = \frac{z-\frac{3}{4}}{\frac{5}{8}}.$$

$$\mathbf{r} = \begin{pmatrix} \frac{3}{4} \\ \frac{5}{2} \\ \frac{3}{4} \end{pmatrix} + \lambda \begin{pmatrix} -\frac{3}{2} \\ -1 \\ \frac{5}{8} \end{pmatrix} \quad \text{or} \quad \mathbf{r} = \begin{pmatrix} \frac{3}{4} \\ \frac{5}{2} \\ \frac{3}{4} \end{pmatrix} + \lambda \begin{pmatrix} -12 \\ -8 \\ 5 \end{pmatrix}.$$

19 g)

$$\mathbf{r} = \begin{pmatrix} -4 \\ 3 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 4 \\ 7 \end{pmatrix}.$$

19 h)

Rewrite as

$$\frac{x-9}{1} = \frac{y-2}{-\frac{5}{2}}, \quad z = 1.$$

$$\mathbf{r} = \begin{pmatrix} 9 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -\frac{5}{2} \\ 0 \end{pmatrix} \quad \text{or} \quad \mathbf{r} = \begin{pmatrix} 9 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ 0 \end{pmatrix}.$$

19 i)

Rewrite as

$$\frac{x}{1} = \frac{z-6}{-1}, \quad y = 5.$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

20 a)

$$\left(\mathbf{r} - \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \right) \times \begin{pmatrix} 5 \\ 2 \\ 6 \end{pmatrix} = \mathbf{0}.$$

20 b)

Rewrite as

$$\frac{x}{9} = \frac{y-2}{-3} = \frac{z-5}{-1}.$$

$$\left(\mathbf{r} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} \right) \times \begin{pmatrix} 9 \\ -3 \\ -1 \end{pmatrix} = \mathbf{0}.$$

20 c)

Rewrite as

$$\frac{x-3}{-1} = \frac{y+1}{1} = \frac{z-3}{-1}.$$

$$\left(\mathbf{r} - \begin{pmatrix} 3 \\ -\frac{1}{2} \\ 3 \end{pmatrix}\right) \times \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}.$$

20 d)

Rewrite as

$$\frac{x+\frac{2}{3}}{\frac{5}{3}} = \frac{y-6}{-1}, \quad z = 2.$$

$$\left(\mathbf{r} - \begin{pmatrix} -\frac{2}{3} \\ 6 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} \frac{5}{3} \\ -1 \\ 0 \end{pmatrix} = \mathbf{0}.$$

20 e)

Rewrite as $x = \frac{1}{3}$,

$$\frac{y-\frac{3}{4}}{\frac{1}{2}} = \frac{z-\frac{5}{3}}{-\frac{5}{3}}.$$

$$\left(\mathbf{r} - \begin{pmatrix} \frac{1}{3} \\ \frac{3}{4} \\ \frac{5}{3} \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ \frac{1}{2} \\ -\frac{5}{3} \end{pmatrix} = \mathbf{0}.$$

20 f)

Rewrite as $x = 4$,

$$\frac{y}{\frac{1}{3}} = \frac{z-2}{-\frac{1}{2}}.$$

$$\left(\mathbf{r} - \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ \frac{1}{3} \\ -\frac{1}{2} \end{pmatrix} = \mathbf{0}.$$